

Top 25 String Problems

● EASY (Must master first – builds foundation)

1. Reverse a String

👉 "hello" → "olleh"

2. Check Palindrome

👉 "madam" → True

3. Count Vowels & Consonants

👉 "hello" → vowels = 2, consonants = 3

4. Remove Spaces from String

👉 "h e l l o" → "hello"

5. Convert to Upper/Lower without built-in

👉 Important logic-based question

6. Count Frequency of Each Character

👉 "aab" → {a:2, b:1}

7. Find Largest Word in Sentence

👉 "I love programming" → "programming"

8. Check Anagram

👉 "listen" & "silent" → True

9. Remove Duplicates from String

👉 "programming" → "progamin"

10. First Non-Repeating Character

👉 "aabcc" → "b"

🟡 MEDIUM (Most asked in placements)

11. Reverse Words in Sentence

👉 "I love Python" → "Python love I"

12. Longest Substring Without Repeating Characters 🔥

👉 "abcabcbb" → "abc"

👉 VERY IMPORTANT (asked in Amazon, etc.)

13. String Compression

👉 "aaabbc" → "a3b2c1"

14. Check Rotation of String

👉 "abcde" & "cdeab" → True

15. Find All Substrings

👉 "abc" → a, b, c, ab, bc, abc

16. Minimum Window Substring 🔥

👉 Advanced sliding window problem

17. Longest Palindromic Substring 🔥

👉 "babad" → "bab"

18. Check Valid Parentheses

👉 "()[]{}" → True

19. Group Anagrams

👉 ["eat","tea","tan"] → [["eat","tea"],["tan"]]

20. Multiply Strings (without int)

👉 "123" * "4" → "492"

🔴 **HARD** (For 12–15 LPA level 🚀)

21. KMP Pattern Matching Algorithm

👉 Efficient string search

22. Rabin-Karp Algorithm

👉 Hash-based searching

23. Z Algorithm

👉 Pattern matching optimization

24. Edit Distance (Levenshtein Distance)

👉 "horse" → "ros" → 3 operations

25. Wildcard Pattern Matching

👉 "a*b?c" type problems

Most IMPORTANT Problems (Don't skip these)

If you're short on time, focus on these 8:

-  Reverse Words
-  Anagram
-  Longest Substring Without Repeating
-  First Non-Repeating Character
-  String Compression
-  Longest Palindrome
-  Valid Parentheses
-  Frequency Count

Pro Tip (THIS is what gets you selected)

Don't just solve — do this:

For each problem:

1. Brute force
2. Optimized solution
3. Explain logic out loud

 Interviewers care about **thinking**, not just code

Reality Check for You

If you can solve:

-  15/25 → You're placement ready for service companies
-  20+/25 → You can target 10–15 LPA roles